

Distance, Speed and Orbits in the Solar System

Javalab Solar System · Javalab

Years 7, 9-10	~30 minutes	Science · Earth & Space Science
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://javalab.org/en/solar_system_en/

Curriculum links: AC9S7U03 · Earth geometry · Earth systems

Learning intentions

- I can compare the relative distances and orbital periods of planets in the solar system.
- I can describe the relationship between a planet's distance from the Sun and its orbital speed.
- I can use a scaled model to represent Earth's position and motion within the solar system.

Getting started

Open Javalab Solar System. Use the controls to speed up time and watch the planets orbit the Sun. Pay attention to the information panel showing each planet's distance from the Sun and its orbital period (how long one orbit takes), and use this to compare planets in the tasks below.

Tasks

1. Speed up time and watch Mercury and Neptune (or the closest and furthest planets shown) complete their orbits. Which one completes an orbit faster? Describe what you see.

2. Record the distance from the Sun and orbital period for five planets, using the simulation's information panel.

Planet	Distance from Sun	Orbital period

3. Looking at your table, describe the general pattern between a planet's distance from the Sun and how long its orbital period is.

4. Explain, using your data, why a planet close to the Sun (like Mercury) completes many more orbits per Earth year than a planet far from the Sun (like Neptune).

5. Locate Earth in the model and note its orbital period. Does this match what you already know about how long a year is on Earth? Explain any difference.

6. Using ideas about gravity and distance (even without formal calculation), suggest a reason why planets further from the Sun tend to move more slowly in their orbits.

7. Compare the spacing between the inner planets (Mercury to Mars) with the spacing between the outer planets (Jupiter to Neptune) in the model. Describe what you notice.

8. Using your table, calculate how many times Mercury orbits the Sun in the time it takes Neptune to complete just one orbit.

Working:

Extension (optional)

If a new dwarf planet were discovered orbiting the Sun at roughly twice Neptune's distance, use the pattern you found in your table to predict whether its orbital period would be longer or much longer than Neptune's, and explain your reasoning.
